Organic Formulae

Exam code: 0620 & 0971

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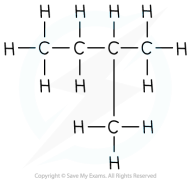
Displayed formulae

- Organic Chemistry is the scientific study of the structure, properties, and reactions of organic compounds.
- Organic compounds are those which contain carbon
- For conventional reasons metal carbonates, carbon dioxide and carbon monoxide are **not** included in organic compounds
- Many of the structures you will be drawing are hydrocarbons
- A **hydrocarbon** is a compound that contains **only** hydrogen and carbon atoms
- Organic compounds can be represented in a number of ways:
  - Displayed Formulae
  - General Formulae
  - Structural Formulae

The **displayed formula** shows the spatial arrangement of all the atoms and bonds in a molecule

- For example:



- This displayed formula tells us several things about the compound
  - It has 5 carbon atoms
  - It has 12 hydrogen atoms
  - It has only single bonds

Help me understand

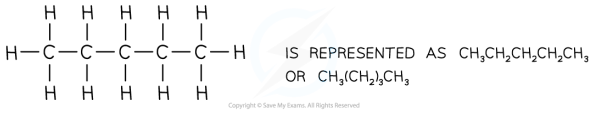
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

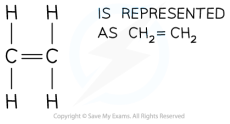
Structural formulae

Extended tier only

- In structural formulae, enough information is shown to make the structure clear, but most of the actual covalent

- Identical groups can be bracketed together
- Side groups are also shown using brackets
- Straight chain alkanes are shown as follows:





Structural isomers

- Structural isomers are compounds that have the **same molecular** formula but **different structural** formulae
- The **molecular formula** is the **actual number** of atoms of each element in a compound
- Compounds with the same molecular formula can have different structural formulae due to the different arrangement of their atoms in space
- Two examples of structural isomers are shown below

Table showing structural isomerism in C_4H_{10}

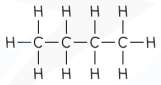
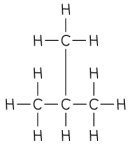
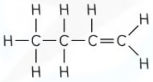
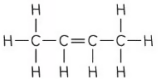
Molecular formula	C_4H_{10}	C_4H_{10}
Structural formula	$CH_3CH_2CH_2CH_3$	$CH_3CH(CH_3)CH_3$
Displayed formula		

Table showing structural isomerism in C_4H_8

Molecular formula	C_4H_8	C_4H_8
Structural formula	$CH_3CH_2CH{=}CH_2$	$CH_3CH{=}CHCH_3$
Displayed formula		

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